Features of the circle

1

a Radius b Diameter c Circumference d Centre

2

a $7\,\text{cm}$ b $5\,\text{cm}$

3 $2x$

4 40 cm

5

a

Diameter (cm)	Radius (cm)
22	11
34	17
38	19
42	21
46	23

b

Diameter (cm)	Radius (cm)
22	11
26	13
28	14
36	18
42	21

Circumference

6

a False b True c True d False

e False f False

7

a
i $13\pi\,\text{cm}$ ii $40.84\,\text{cm}$

b
i $36\pi\,\text{cm}$ ii $113.10\,\text{cm}$

c
i $7.7\pi\,\text{cm}$ ii $24.19\,\text{cm}$

d
i $8.2\pi\,\text{cm}$ ii $25.76\,\text{cm}$

8

a $59\pi\,\text{cm}$ b $44\pi\,\text{cm}$

9

10

a 10π cm

b 10π cm

c They got the same result because the diameter is always equal to twice the radius, so πd and $2\pi r$ are always equal.

11

a $\dfrac{26}{\pi}$ cm

b 8.28 cm

12

a 6 cm

b $\dfrac{6}{\pi}$ cm

c 16 cm

d $\dfrac{13}{\pi}$ cm

13

a 2.23 cm

b 2.86 cm

c 2.37 cm

d 15.60 cm

14

a $\dfrac{36}{\pi}$ cm

b 11.46 cm

15

a 18 cm

b $\dfrac{18}{\pi}$ cm

c 17 cm

d $\dfrac{14}{\pi}$ cm

16

a 11.78 cm

b 14.01 cm

c 22.28 cm

d 7.80 cm

17

Radius (cm)	Diameter (cm)	Circumference (cm)
2	4	4π
8	16	16π
41	82	82π
$\dfrac{170}{\pi}$	$\dfrac{340}{\pi}$	340

18

a 13.37 m

b 6.68 m

Applications

19 100.5 cm

20 106.8 cm

21 25.1 m

22 94.2 mm

23 311.0 cm

24 22.1 laps

25 $39\,788$ revolutions

26 345.6 m

27 163.4 m

28 8

29 879.6 m

30 231.0 km